

ST. XAVIER SENIOR SECONDARY SCHOOL

ANNUAL EXAMINATION - 2026

Subject: Science & Mathematics (Class 10) Max Marks: 80 Time: 3 Hours

Q1. Find the roots of the quadratic equation $ax^2 + bx + c = 0$ using the quadratic formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$. [1]

(A) $x = \frac{-b \pm \sqrt{D}}{2a}$

(B) $x = \frac{-b \pm D}{2a}$

(C) $x = \frac{b \pm \sqrt{D}}{a}$

(D) $x = -b / (2a)$

Q2. Balance the following chemical reaction: $2H_2 + O_2 \rightarrow 2H_2O$. Name the precipitate formed when $BaCl_2$ reacts with H_2S . [1]

(A) $BaSO_4$ (White Precipitate)

(B) HCl gas

(C) BaO

(D) SO_2

Q3. An electron with charge $q = 1.6 \times 10^{-19} \text{ C}$ moves in a magnetic field. State Newton's second law $F = ma$ and calculate the force acting on it. [1]

(A) 10 N

(B) 2.5 N

(C) 7 N

(D) 20 N

